

Тригонометрические многочлены и экстремальные задачи

1 сентября 2026 года

Теоретический минимум.

Опр. A trigonometric polynomial of degree n is $T(\theta) = a_0 + \sum_{k=1}^n (a_k \cos k\theta + b_k \sin k\theta)$; equivalently $T(\theta) = \sum_{|k| \leq n} c_k e^{ik\theta}$ with $c_{-k} = \overline{c_k}$.

Теорема 1 (о нулях) A nonzero trigonometric polynomial of degree n has at most $2n$ zeros on $[0, 2\pi)$ counted with multiplicity; hence it is determined by its values at any $2n + 1$ points.

Теорема 2 (Fejér–Riesz 1916) A trigonometric polynomial $T \geq 0$ of degree n can be written as $T(\theta) = |p(e^{i\theta})|^2$ for an algebraic polynomial p of degree n ; in particular T is a square in the algebra of trigonometric polynomials of the same degree.

Факт 1 The Fejér kernel $K_n(\theta) = \frac{1}{n+1} \left(\frac{\sin \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}} \right)^2$ is non-negative, has integral 1 and concentrates at 0; Cesàro means of Fourier series are convolutions with it, whence Fejér's theorem.

1. Prove Theorem 1 (both by passing to an algebraic polynomial in $z = e^{i\theta}$ and by a Rolle argument), and deduce that a trigonometric polynomial of degree n vanishing at $2n + 1$ points is identically zero.
2. Prove that the Fejér kernel is non-negative and has integral 1, and deduce Fejér's theorem: the Cesàro means of the Fourier series of a continuous 2π -periodic function converge to it uniformly. Deduce the Weierstrass approximation theorem for trigonometric polynomials.
3. (Putnam 2013) Let $\mathcal{C}$ be the set of cosine polynomials $f(x) = 1 + \sum_{n=1}^N a_n \cos(2\pi nx)$ such that $f(x) \geq 0$ for all real x and $a_n = 0$ whenever n is a multiple of 3. Determine the maximum value of $f(0)$ over $f \in \mathcal{C}$ and prove that it is attained.

4. (Proposed problems, №271) Prove that $\sum_{n=0}^{\infty} \frac{\cos n\theta}{(n+1)^4} \geq \frac{7\pi^4}{720}$ for every real θ , and find the point where equality is approached.
5. Prove the Fejér–Riesz theorem, and use it to show that a non-negative trigonometric polynomial of degree n with $a_0 = 1$ satisfies $|a_k| \leq 2$ for all k .